

Bhiman Kumar Baghel

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Professional Summary

Ph.D. researcher and former Samsung Lead NLP Engineer with 8+ years of experience developing AI systems from research to production. My work focuses on efficient and reliable reasoning in language models and AI agents, including training-free tool-use adaptation and model editing. Published 6 papers, hold 4 patents, and shipped conversational AI to 100M+ devices. Seeking full-time Research Scientist and Applied Scientist roles focused on building capable, efficient, and deployable AI systems.

Skills

ML / LLM Frameworks: PyTorch, Hugging Face Transformers, PEFT / LoRA, vLLM, PyTorch Lightning, TensorFlow

Cloud & MLOps: Amazon Bedrock, AWS (EC2, S3), Docker, MLflow, Weights & Biases, Git + CI/CD

Languages & Databases: Python, C++, Pandas, NumPy, SQL, MongoDB, Neo4j

Education

University of Pittsburgh, PA, USA

August 2023 – April 2027

PhD, Computer Science (GPA: 3.655/4)

Indian Institute Of Technology, Kharagpur, India

July 2017 – May 2019

Master's, Computer Science (GPA: 8.82/10)

Research & Industry Experience

Applied Scientist II Intern, Amazon – Seattle, WA, USA

May 2026 - Present

- Developing enterprise autonomous agents for complex enterprise workflows.

Applied Scientist II Intern, Amazon – Seattle, WA, USA

October 2025 - January 2026

- Developed a training-free framework to improve tool-use reasoning in AI agents by extracting compositional rules from agent failures via single-pass meta-reasoning and distilling them into inference prompts across 78 API domains, enabling resource-constrained deployment without model training or iterative rollouts.
- Enabled Qwen3-30B to outperform the best Claude-4.5-Sonnet configuration by 15.7% with 5× faster inference; surpassed a state-of-the-art context-engineering method by 38.8% at 4.1× lower cost.

Graduate Research Assistant, University Of Pittsburgh – PA, USA

August 2024 – Sept 2025

- Engineered a plug-and-play iterative editing pipeline that enhanced edit-success rate by 38 percentage points over prior SOTA on LLaMA-3/2 and GPT-J, enabling rapid knowledge updates without full-model fine-tuning.
- Conducted a gender-bias audit of GPT-3.5 and BART summaries over 19,579 student reflections; used Jensen–Shannon divergence to reveal a 10% male-topic skew and uncovered under-represented female topics.
- Built a 2,900-meme multimodal dataset; my manual audit revealed stereotype bias in 40% of LLaVA and MiniGPT-4 explanations, traced to visual/named-entity stereotypes, and text–image representation imbalance.

Lead NLP Engineer, Samsung Research – Bangalore, India

June 2019 – August 2023

- Spearheaded CoSMIC, a BERT-based multi-intent NLU engine for SmartThings; shipped to 100 M + devices, reaching 96% intent accuracy and cutting live NLU errors by 67%.
- Localized and scaled CoSMIC for the Korean market, mentoring a cross-site team and re-engineering tokenization to lift intent-slot F_1 by 25%.
- Architected production conversational-AI models (intent, slot, OOD) that raised multi-intent F_1 from 87% → 92% and achieved 90% OOD recall across all public benchmarks.

Machine Learning Intern, IBM – Bangalore, India

May 2018 – July 2018

- Prototyped an LSTM-based anomaly-prediction engine that monitors 33 infrastructure health metrics and launches auto-remediation scripts, forecasting critical failures with 97% precision.

Publications

- Resolving UnderEdit & OverEdit with Iterative & Neighbor-Assisted Model Editing** EMNLP Findings (2025)
Bhiman Kumar Baghel, Emma Jordan, Zheyuan Ryan Shi, Xiang Lorraine [PDF]
- CreativityPrism: A Holistic Evaluation Framework for Large Language Model Creativity** arXiv (2026)
Zhaoyi Joey Hou, Bowei Alvin Zhang, Yining Lu, *Bhiman Kumar Baghel*, Anneliese Brei, Ximing Lu, Meng Jiang, Faeze Brahman, Snigdha Chaturvedi, Haw-Shiuan Chang, Daniel Khashabi, Xiang Lorraine Li [PDF]
- A Fairness Analysis of Human and AI-Generated Student Reflection Summaries** GeBNLP Workshop, ACL (2024)
Bhiman Kumar Baghel, Arun Balajiee Lekshmi Narayanan, Michael Miller Yoder [PDF] [Talk]
- Multimodal Understanding of Memes with Fair Explanations** MULA Workshop, CVPR (2024)
Yang Zhong, *Bhiman Kumar Baghel* [PDF] [Talk]
- Intent-Focused Semantic Parsing and Zero-Shot Learning for Out-of-Domain Detection in Spoken Language Understanding** IEEE Access (2021)
Niraj Kumar, *Bhiman Kumar Baghel* [PDF]
- Smart Stacking of Deep Learning Models for Granular Joint Intent-Slot Extraction for Multi-Intent SLU** IEEE Access (2021)
Niraj Kumar, *Bhiman Kumar Baghel* [PDF]

Patents

- Method and system for time-based personalization management in multi-device environment** WO2025018568A1 (2024)
Sourabh Tiwari, *Bhiman Kumar Baghel*, Jalaj Sharma, Manish Chauhan, Boddu Venkata Krishna Vinay, Syed Khaja Moinuddin [Link]
- Methods and systems for enabling seamless indirect interactions** US18517995 (2023)
Venkata Krishna Boddu Vinay, *Bhiman Kumar Baghel*, Gorang Maniar, Syed Khaja Moinuddin, Sudhansu Ranjan Acharya [Link]
- Method and system for mitigating physical risks in an IoT environment** US18202687 (2023)
Niraj Kumar, *Bhiman Kumar Baghel* [Link]
- Methods and systems for determining missing slots associated with a voice command for an advanced voice interaction** US17835387 (2023)
Niraj Kumar, *Bhiman Kumar Baghel* [Link]

Honors & Awards

- Samsung High Performance Bonus** (3×), Samsung Research – Bangalore, India 2023
- Samsung Excellence Award** (5×), Samsung Research – Bangalore, India 2023
Recognized for SmartThings CLab innovation finalist and 4 US A1 patent filings.
- 2nd Runner-Up, Audience Poll**, IBM Extreme Blue Expo – Bangalore, India 2018
Voted top-3 of 24 projects by 100+ expo attendees.

Academic Service

- Program Committee Member**, Explainable Automated Software Engineering (ExASE) Workshop, ASE 2025
- Reviewer**, Multimodal Learning and Applications (MULA) Workshop, CVPR 2025
- Reviewer**, Gender Bias in NLP (GeBNLP) Workshop, ACL 2025
- Reviewer**, Representation Learning for NLP (RepL4NLP) Workshop, NAACL 2025